

Sloppy - Testing Prompt (#258239)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

Pre-registered on:

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1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

H1: In a cue-learning task in which none of the cues has an influence on the outcome, people who learned in the presence of a positive (vs. flat) exogenous trend will be more likely to believe that the (non-influential) cues had an impact on the outcome.

H2: Prompting participants to embrace a negative-test strategy (versus a positive-test strategy, or no strategy in particular) will attenuate the impact of the exogenous trend on illusory beliefs.

3) Describe the key dependent variable(s) specifying how they will be measured.

DV1: Subjective understanding. Participants will report their subjective understanding of the relationship between cues and outcomes on three 5-point bipolar scale items. These will be averaged into a composite measure of subjective understanding.

DV2: Endorsement of illusory correlations. Participants will see four statements, each claiming that a non-diagnostic cue had an impact on the outcome. They will be asked to report their belief that these false statements are correct on 5-point scale items anchored at Not at all confident (1) and Extremely confident (5). These four items will be averaged into a composite measure of endorsement of illusory correlations.

We will measure both DVs twice: After fourteen rounds of the game (the "Interim" measures: SubjUnderstanding_Interim; IllusoryCorr_Interim) and at the end of the twenty rounds of the game (the "Final" measures: SubjUnderstanding_Final; IllusoryCorr_Final).

4) How many and which conditions will participants be assigned to?

Participants will be randomly assigned to one of six cells of a 2 (Slope: Increasing vs. Flat) by 3 (Testing Prompt: Positive vs. Negative vs. Control) design.

The Slope factor will manipulate the slope of the payoffs over the twenty rounds of the game. The Testing Prompt factor manipulates the kind of testing prompt participants will receive after playing 14 rounds of the game: One that encourages a "positive-test strategy", a "negative-test strategy", or a "control strategy".

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will first analyze the Interim measures using a t-test, comparing participants in the "Positive" and "Neutral" Slope condition. At this stage, participants are not randomized to a Testing Prompt condition yet, so incorporating this factor is not required.

We will then analyze the Final measures using the following ANCOVA model:

Final = Slope + Testing_Prompt + Slope x Testing_Prompt + Interim.

This model provides the greatest amount of statistical power (compared to a Final – Interim difference score) to identify changes in confidence induced by different testing prompts.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will measure the amount of time participants spend on the cue-learning task, and exclude all participants who completed the task in less than 110 seconds.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will collect 1000 observations. Participants excluded for completing the task in less than 110 seconds will not be recollected.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will measure, at the end of the experiment, whether participants believed that the outcomes increased (vs. stayed flat vs. decreased) over time.

We will also measure how often participants relied on a positive (vs. negative) test strategy in the first half (vs. the second half) of the game.

We will report both variables as a descriptive, exploratory analysis.